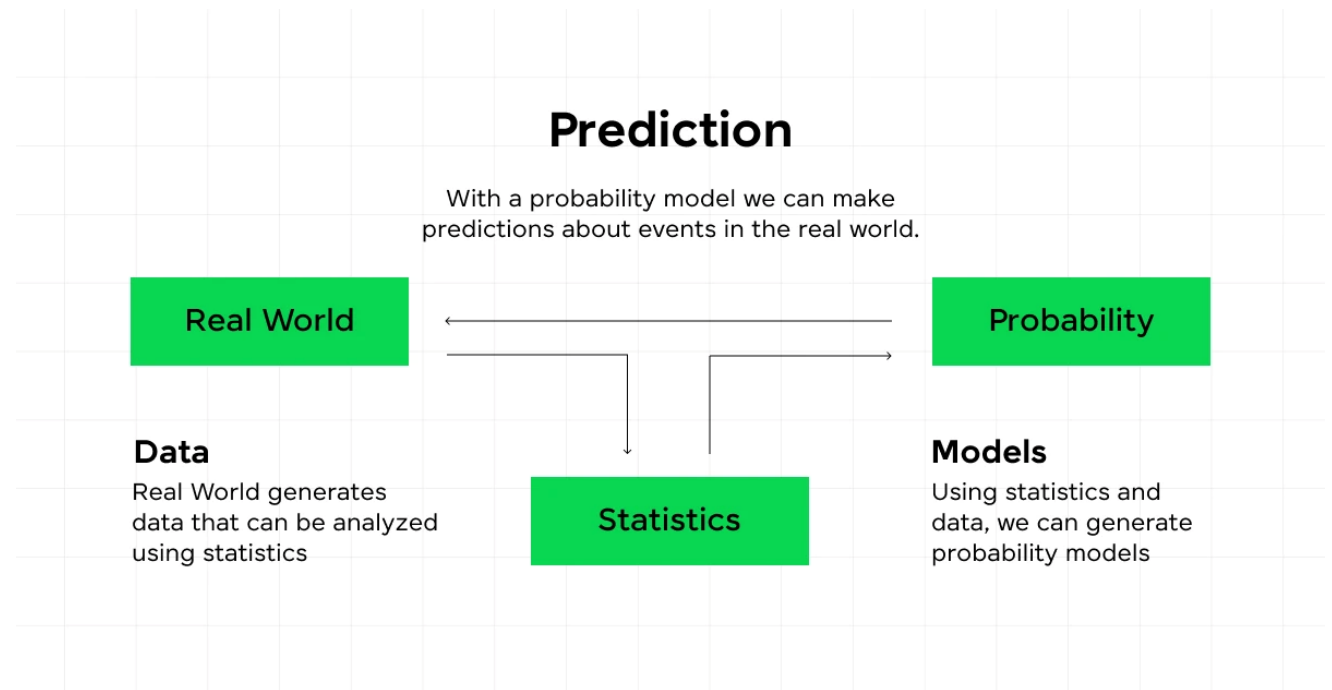
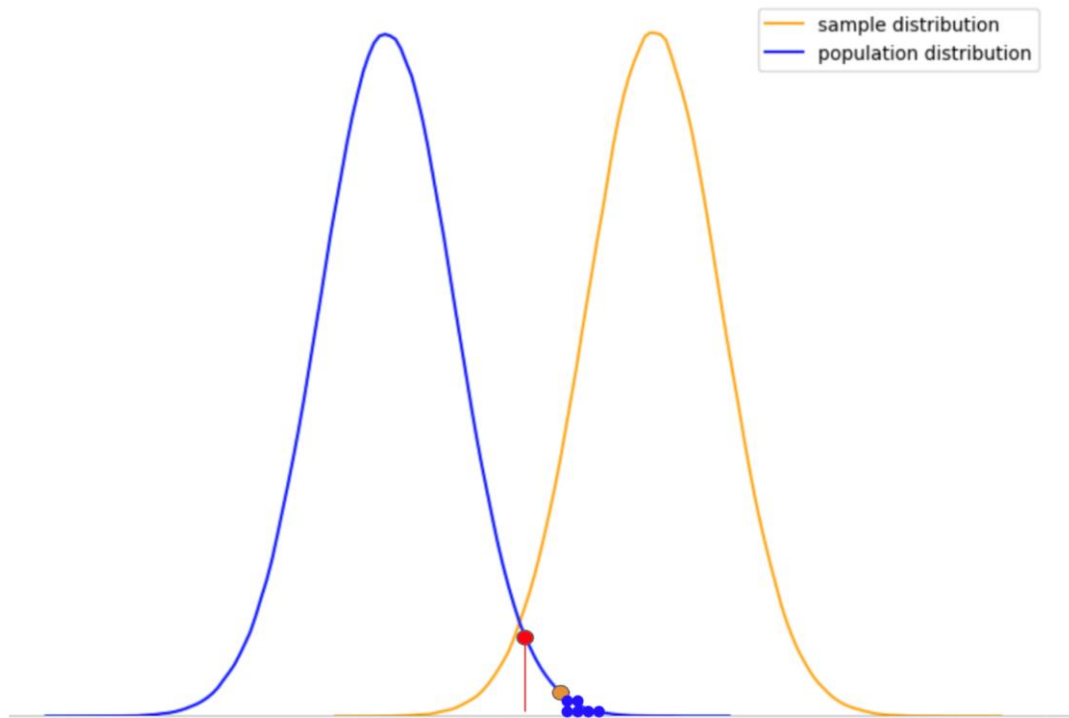

The Shape of Probability

- **Normal Distribution:** The "Bell Curve." Most natural data (like human heights or exam scores) clusters around the middle.
- **Binomial Distribution:** Used for "Yes/No" outcomes (e.g., Did a patient recover? Did a seed sprout?). It models the number of successes in a fixed number of trials.
- **Poisson Distribution:** Models the frequency of events over a fixed period of time or space (e.g., How many cars pass a Charsadda checkpoint per hour?).

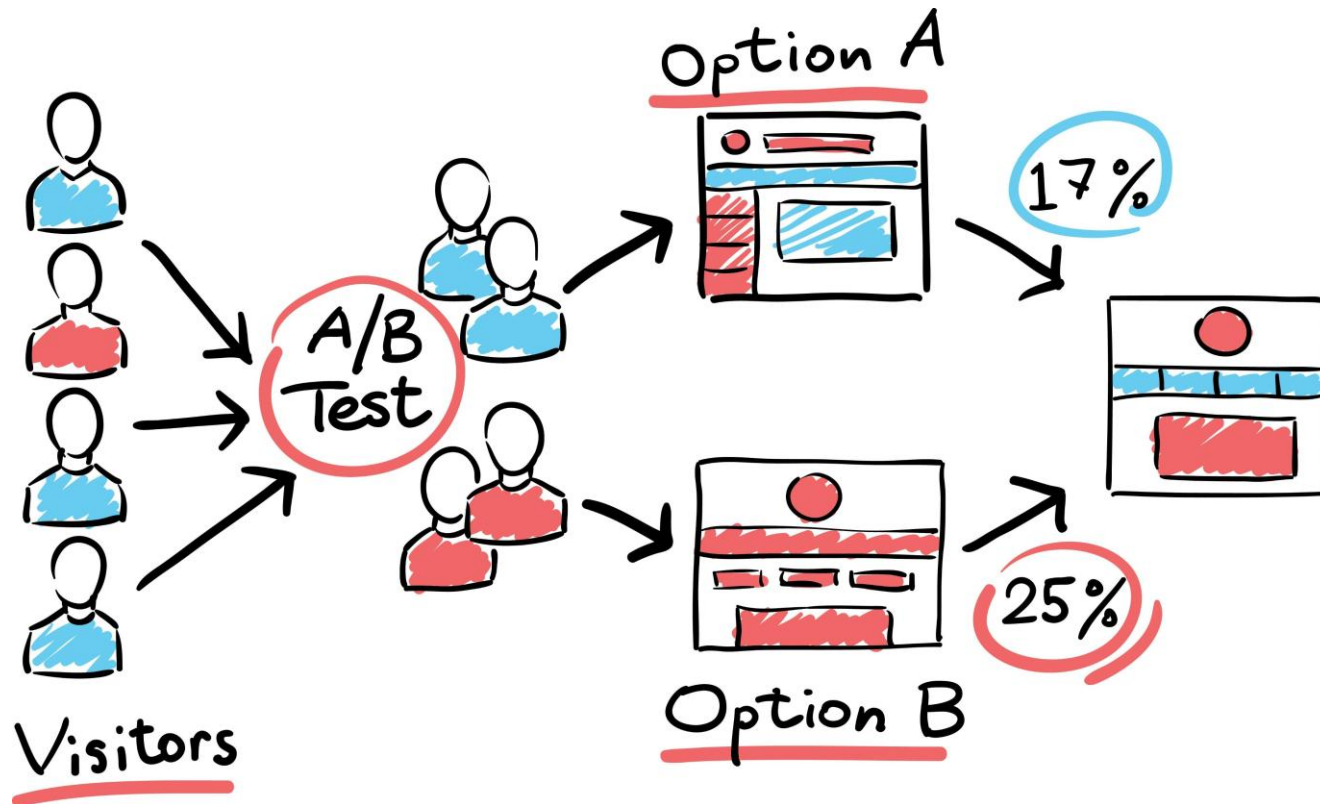


The P-Value – The "Luck" Detector



- **The Null Hypothesis (H_0):** The assumption that "Nothing happened" or "The difference is just luck."
- **Alpha Level (α):** Usually set at **0.05**. This is our "Doubt Threshold."
- **The P-Value:** The probability that we would see these results if the Null Hypothesis were true.
- **The Rule:** If $P < 0.05$, we "Reject the Null." We are 95% confident the result is a real trend, not a mistake.

A/B Testing – Experimental Design



- **The Setup:** Splitting a population into two groups: **Control** (A) and **Treatment** (B).
- **The Change:** Group B gets a "treatment" (a new medicine, a different fertilizer, or a new website button).
- **The Measurement:** We use statistical tests to see if the difference in performance between A and B is large enough to be meaningful.
- **Industry Standard:** This is how companies like Facebook, Google, and Amazon make every single design decision.